Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/01/2005
PI Name:	Green, Lora Ph.D.		
Project Title:	Response of Thyroid Functional Tissue Units to Space-Like Radiation Fields		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92354	**Congressional District:**	41
Comments:	PI is deceased as of January 2010.		
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
Start Date:	02/12/2004	**End Date:**	04/01/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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The thyroid like most epithelial-based glandular tissues are a common site of tumor development and sensitive to ionizing radiation induced carcinogenic changes. Using thyroid as a model, especially the well-defined normal cell line that we have characterized, provides us the opportunity to investigate how normal tissue responds to ionizing radiation from different sources, doses and dose rates, at multiple levels. From “Radiation Hazards to Crews of Interplanetary Missions” a major question raised was “Are there studies that can be conducted to increase the confidence of extrapolation from rodents to humans of radiation-induced genetic alterations that in turn could enhance similar extrapolations for cancer?” We have established tissue analogs that we believe will increase the certainty of this type of extrapolation. The thyroid tissue analog produced during growth in a low-shear bioreactor is a realistic three-dimensional replica of the tissue being modeled. The bioreactor system provides stable environmental controls not achievable by other methods. Thus, our goal is to confirm and extend the characteristic pattern of radiation modulated

Task Description:	gene expression in rat thyroid tissue analogs using space-like low dose/low dose rate radiation, and challenge the chronically irradiated tissue analogs. To- Confirm and extend the characterization of the pattern of radiation modulated gene expression in rat thyroid tissue analogs using space-like low-dose/low-dose rate radiation of low, medium and high linear energy transfer (LET). We hypothesized that in the LEO analog a low-dose/low-dose rate “priming” exposure to radiation would invoke a suite of cell and tissue level homeostatic mechanisms that would be reflected in gene expression profiles and complementary structural and functional changes in component cells, as well as in their functional tissue units (FTU’s, i.e. follicles). Once the thyrocytes have completed the “transition” to the “primed” state, the thyroid FTU’s will have an altered response to an acute challenge exposure to radiation that will take the form of increased resistance to biological changes relative to “unprimed” cells. The response of the FTU’s to the GCR exposures is less predictable as no information comparable to our system is available. However, based on our previous work we expect that principal regulatory elements in the priming transition will include transforming growth factor-beta and protein kinase C isoforms. These important products will direct other respondent genes that will be identified by microarray expression profiling. It is expected that the transcription factors, oncogenes and cell coordinating gene products will exhibit the most significant alterations. The tissue analogs will be subjected to various dose and delivery schedules, of gamma rays and protons (LLU), as well as carbon and iron particles (NSRL, BNL), to obtain gene expression profiles that reflect critical changes in structure and functional aspects of thyroid homeostasis, or normalcy. Generally these studies address the following -- Thyroid tolerance, radiation dose limit to head & neck, susceptibility of cancer in normal tissue. Additionally these studies address several points indicated in the “Strategic Proposed Plan for Space Radiation Health Research” (1998). What mechanisms are involved in modulating radiation damage at the molecular level (repair enzymes, etc.)? What is the probability of initiating neoplastic cell transformation or other steps leading to a cancerous cell? How do cellular repair mechanisms modulate damage produced by ionizing radiation? And how can the radiation effects on cells in culture be related to radiation effects in “normal cells”?
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The majority of responses we have measured in the bioreactor generated tissues are those related to tissues, extracellular matrix and rather than cell specific products. This supports the premise of the proposal that the more highly-organized tissues better represent the in vivo condition than we would have obtained using two-dimensional flasks. This relates to NASA’s critical question regarding the ability to extrapolate between in vitro and in vivo models. Our data suggests that the more organized tissues of the bioreactor provide an advantage that allows for an in vitro model to better mimic the in vivo condition, without the added costs of animal. There have been several interesting results in the preliminary stages of analyzing the data. The most pronounced finding to date has been that the three-dimensionally organized tissue analogs are less responsive than traditional flask grown cultures. This implies that the data obtained in vitro using cell lines that are two-dimensionally organized may overestimate the sensitivity of particular cell types. That is that tissue and in deed whole bodies are more resistant to the effects of radiation that would be predicted from standard tissue culture. This is even more exaggerated if the cells are transformed rather than from a normal source.
Bibliography Type:	Description: (Last Updated: 01/29/2015)
Abstracts for Journals and Proceedings	Green LM, Kyeyune-Nyombi E, Ortloff L, Rightnar SS & Bianski BM "Response of Thyroid Tissue Units to Space-Like Radiation Fields: Comparison of Acute Low-Dose Exposures of Differing LET" 3rd International Workshop on Space Radiation and 15th Annual NASA Space Radiation Health Investigators’ Workshop, Port Jefferson, New York May 16-20, 2004 , May-2004
Abstracts for Journals and Proceedings	Green LM, Bianski BM, Ortloff LS, Kyeyune-Nyombi E & Rightnar SS "Global Assessment of Gene Expression in Thyroid Tissue Analogs Exposed to Space-Like Radiation." Bioastronautics Investigators Workshop Galveston Texas , Jan-2005